STAT 547 - Data Mining
Homework 1

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Part 1: The Nine Wheels Framework

Question 1: The Wheel of Problem Formulation

A healthcare company wants to predict patient readmission within 30 days.

(a) Define this problem using the Nine Wheels terminology: $\mathcal{X}$ (input space), $\mathcal{Y}$ (output space), $\mathcal{H}$ (hypothesis space), and ℓ (loss function).

Broadly defined, a hospital readmission is when a patient who had been discharged from a hospital is admitted again to that hospital or another hospital within a specified time frame.

Let the input space $\mathcal{X}$ contain variables that the healthcare or hospital has on the patients. This can be a p dimension vector of predictor variables such as patient's demographic, age, height, historical diagnostic data, number of readmission in the past, conditions of patient at discharge date and medical measurements. The output space $\mathcal{Y}$ should be $\{0, 1\}$, 0 for no readmission and 1 for readmission within 30 days. It is clear that the output space is this way as the healthcare company wants to predict wether a patient is readmitted in 30 days or not. Our hypothesis space $\mathcal{H}$ will be defined by the set of functions that maps the input space $\mathcal{X} \to \mathcal{Y}$ under a linear combination of our predictor variables discussed in $\mathcal{X}$.

$$\mathcal{H} = \left\{ f_{\mathcal{H}}(x) = \left(1 + \exp(-(w^T x + b))\right)^{-1} \middle| w \in \mathbb{R}^p, b \in \mathbb{R} \right\} \tag{1}$$

It follows that we use the Binary Cross-Entropy as our loss function

$$\ell(y, \hat{p}) = -\left(y \log \hat{p} + (1 - y) \log(1 - \hat{p})\right), \quad \hat{p} = f_{\mathcal{H}}(x) \in \mathcal{H} \tag{2}$$

This loss function make sense intuitively as the negative log function is larger for lower $\hat{p}$ which acts like a "penalizer" because the optimal $f^* \in \mathcal{H}$ is reached from minimizing ℓ. Also we take the negative log as our domain for $\hat{p} \in [0, 1]$ due to the sigmoid function.

$$f^* = \arg\min_{\hat{p}} \ell(y, \hat{p}) \tag{3}$$

(b) Explain why precision might be more important than accuracy here, using the business impact framework from Wheel 9.

We define precision and accuracy in the respective order.

$$\text{Precision} = \mathbb{P}(y = 1 | \hat{f} = 1\}$$

$$\text{Accuracy} = \mathbb{P}(y = \hat{f})$$

We want to have high precision due to the fact we rather have not miss any patients for readmission. It is especially important to classify a patient as a candidate for readmission when they truly need it. Also note that it is less costly for the hospital to let go of a patient as classify the patient as a candidate for readmission even when they are truly not, since the hospital will be fined and the patient's family would most likely want to be safe than sorry. On the other hand, accuracy focus not just on the positives being classify as positive but also classifying the patient as negative when they are actually negative too. This is certainly helpful but less important, because even if we classify a true negative as positive it is less "costly" in this context.

(c) Describe what data veracity ($\mathbb{1}\{\text{data is reliable}\}$) means in healthcare and three specific challenges.

Data veracity in healthcare means that the data from the patients is reliable for the models to operate on, and is able to spit our results that is significant and accurate. Possible challenges include and are not limited to

- Missing Data: When a certain variable is not recorded from a patient such as glucose level and such.
- Noise: There is possible random error from data entry and machine measurements from the medical equipments.
- Biases: The data may not always be i.i.d as patients of a certain hospital may only represent patient of a certain region but not enough for the health company.

Question 2: The Wheel of Theoretical Guarantees

Consider the k-Nearest Neighbors (k-NN) algorithm for classification.

(a) Define the Bayes classifier $f^*(x)$ and its associated Bayes risk. Why is it the theoretical gold standard? The Bayes classifier is defined as the the predictor that achieves the minimum possible risk over all possible functions in the universal space $\mathcal{Y}^{\mathcal{X}}$. This is the power set all sets of possibilities and the theoretical f_B^* is the Bayes classifier. Under the 0-1 loss $\ell_{0/1}$ the Bayes classifier is

$$f_B^* = \arg\max_{y \in \mathcal{Y}} \mathbb{P}(Y = y|X = x) \tag{4}$$

This is the theoretical gold standard because if there were any other predictor models f_o^*, then the excess risk will always be positive because f_B^* is the optimal classifier in the entire universe.

$$R(f_o^*) - R(f_B^*) > 0, \quad R(\cdot) = \mathbb{E}[\ell(y, \cdot)] \tag{5}$$

However we know in application in all cases it is almost computationally impossible or inefficient to search the entire space of $\mathcal{Y}^{\mathcal{X}}$ for this mythical f_B^*.

(b) State the universal consistency result for k-NN. What conditions on k and n are required for the k-NN risk to converge to the Bayes risk?

There are three usual conditions of k and n so that $R(f_{KNN}^*) \to R(f_B^*)$. Note that similar to the ideas of a epsilon ball in point set topology, for a new data point $x_0 \in \mathcal{X}$ we rather define a $\mathcal{V}_k(x_0)$,

$$\mathcal{V}_k(x_0) = \{x_i \in \mathcal{D}_n | d(x_0, x_i) < d_{(k)}, i \in \mathbb{N}\} \tag{6}$$

So it follows that if we have theoretically a continuous growing n (e.g $n \to \infty$) then obtain a extremely dense space that can represent our features or predictor variables, then $k \to \infty$ would allow us to have more a bigger $\mathcal{V}_{\|}$ ball/neighborhood to catch up with the growing amount of samples. However there must be a constrain on k such that $n >> k$ so $\frac{k}{n} \to 0$ because it would not make sense to have more neighbors than the allowable sample size n. In this context if k is too big then the metrics $d_{(i)}$'s is almost useless as it may be measuring how close x_0 is with something very far away. Thus we need n to grow faster or be much larger than k to keep the neighbors of x_0 truly "neighbors".

$$\lim_{n,k \to \infty} \mathbb{E}[R(f_{KNN}^*(n, k)] = R(f_B^*), \quad n >> k \tag{7}$$

$$\frac{1}{k} \sum_{i \in \mathcal{V}_k(x_0)} \mathbb{1}\{y_i = 1\} \xrightarrow{P} \mathbb{P}(Y = 1|x_0) \tag{8}$$

The above is achieved when the three conditions is well maintained. *A Remark: As the number of features p grows we can see that almost all data points $x \in \mathcal{X}$ is considered (i.e $k \approx n$). For a new data point x_0, k-NN will calculate all $d_{(k)}$ distances with x_0. Then the question becomes what is the most popular group within the data, and f_{k-NN} will most likely point x_0 to be classified as the most frequent group in the data.*

(c) Connect this result to the bias-variance tradeoff. How does the choice of k control this tradeoff in practice, even for finite n?

For a finite sample size n if k is small then there will be very high variance in the model as there is a lot of variations in the predictions. Say for the smallest possible k-NN we have a 1-NN, then it is plausible that if for a small change in the "location" of the x_0 we can get very different results based on the group of closest neighbor to x_0. The bias on the other hand for this case very low as the 1-NN model is very complexed and fits the neighborhood $\mathcal{V}_1$ perfectly around x_0 excluding all other possible points in $\mathcal{X}$.

Part 2: k-Nearest Neighbors Learning

Question 3: The Distance Perspective

(a) The performance of k-NN is highly sensitive to the choice of distance metric. For a feature vector $x = (\text{age}, \text{weight}, \text{cholesterol})$, propose and justify a specific distance metric. Assume age is in years, weight in kg, and cholesterol in mg/dL.

It is clear that using the raw form of this feature vector x is not a very good idea as a distance of 10 on age is very different from the distance of 10 for the weight variable for a normal Euclidean metric. We choose to normalize our data with standardization so that we can remove the unit biases in the data. If call our standardized vector variables $\tilde{x} = (\tilde{a}, \tilde{w}, \tilde{c})$, then

$$\tilde{x} = \left(\frac{a - \frac{1}{n}\sum_{n \geq i} a_i}{\frac{1}{n-1}\sqrt{\sum_{i \leq n}(a_i - \frac{1}{n}\sum_{n \geq i} a_i)^2}}, \frac{w - \frac{1}{n}\sum_{n \geq i} w_i}{\frac{1}{n-1}\sqrt{\sum_{i \leq n}(w_i - \frac{1}{n}\sum_{n \geq i} w_i)^2}}, \frac{c - \frac{1}{n}\sum_{n \geq i} c_i}{\frac{1}{n-1}\sqrt{\sum_{i \leq n}(c_i - \frac{1}{n}\sum_{n \geq i})^2}} \right) \tag{9}$$

Since each feature will still be a real valued number after this standardization, we can say that the euclidean metric space holds from properties of the real numbers.

(b) Explain the concept of the curse of dimensionality in the context of k-NN. How does the distribution of nearest-neighbor distances change as the number of features p grows?

In the context of k-NN, the curse of dimensionality refers to the fact that as p increases, the feature space becomes exponentially sparse because volume grows at a some exponential rate of r^p. Then this will forces the k-NN algorithm to look across a much larger "volume" to find its k neighbors. Also note that even if it does find a neighbor it may actually not be a close neighbor in high dimensions under the Euclidean metric.

(c) Describe two preprocessing techniques (from Wheel 3) that are essential for effective k-NN learning and explain why.

One is similar to what we have already shown above is centering which creates a centered data matrix $\overline{X}$. This is constructed by subtracting the sample mean vector $\overline{x}$ from each of the original data vectors x_i. The second one would be normalizing which is exactly what we have shown in part (3a). In this context, if we do not scale or normalize, the feature with the largest numerical range (e.g., cholesterol) would dominate the distance calculation, making the algorithm treat a minor clinical change in cholesterol as more significant than a massive demographic change in age.

Question 4: The Model Complexity Perspective

(a) In k-NN classification, how does the value of k directly control the complexity of the hypothesis space $\mathcal{H}$? Is a 1-NN model a high-capacity or low-capacity model? Justify your answer.

(b) Sketch two plots: (i) model complexity (y-axis) vs. k (x-axis), and (ii) expected prediction error (y-axis) vs. k (x-axis). Label the regions of underfitting and overfitting.

(c) Contrast k-NN (a lazy, instance-based learner) with a parametric model like logistic regression (an eager, model-based learner). List one significant advantage and one significant disadvantage of each paradigm.

Part 3: Contrasting Learning Paradigms

Question 5: Frequentist Gaussian Discriminant Analysis

Consider Gaussian Discriminant Analysis (GDA) for a two-class classification problem, C_1 and C_2, with class-conditional densities $p(x|C_k) = \mathcal{N}(x|\mu_k, \Sigma)$.

(a) Derive the log-odds $\log\left(\frac{P(C_1|x)}{P(C_2|x)}\right)$ in terms of the model parameters $(\mu_1, \mu_2, \Sigma, \pi_1, \pi_2)$. Show that the decision boundary is a linear function of x.

(b) Write down the closed-form expressions for the Maximum Likelihood Estimates (MLE) of the parameters μ_1, μ_2, and the shared covariance matrix Σ.

(c) Interpret the MLE for Σ. How does it pool information from both classes, and what is a key statistical advantage of this pooling compared to having separate covariances?

Question 6: Computational Complexity and Scalability

(a) Analyze the computational complexity of the standard k-NN algorithm for making a single prediction. Break it down into the complexity of (i) calculating distances and (ii) finding the k smallest distances.

(b) The standard k-NN algorithm has a time complexity of $O(nd)$ for a single prediction, where n is the training set size and d is the number of dimensions. Explain why this does not scale well for real-time applications with large n and d.

(c) Research and describe one algorithmic strategy (e.g., KD-Trees, Ball Trees, approximate nearest neighbors) used to accelerate the nearest neighbor search. Explain its core idea and under what conditions it is most effective.

Part 4: Computational Demonstration

Question 7: Simulation Insights

(a) Based on your implementation of the k-NN bias-variance tradeoff simulation, what does the visualization teach about the tradeoff? How does this connect to Question 4?

(b) After contrasting k-NN and GDA on a simple dataset, compare the decision boundaries. What does this tell you about the inherent assumptions of k-NN vs. GDA?

Question 8: Video Presentation

(a) Provide a link to your 5-minute video presentation covering: one surprising insight from Questions 1-6, the connection between k-NN theoretical guarantees and the curse of dimensionality, your personal definition of "AI-aware" statistics, and one proposal for course improvement.

Part 5: Idiosyncratic Reflection (Mandatory)

Question 9: Self-Assessment
(a) In your own words, explain the concepts from the first few weeks of STAT 547 that you feel you have truly mastered. What is the most challenging concept or practical skill that you are still working to solidify?
(b) After completing this homework assignment, what grade do you estimate you will earn? Justify your estimate by referencing which exercises you found straightforward and which you found most challenging.